Laporan Praktikum #1

Mata Kuliah	:	Teknologi Big Data
Dosen Pengampu	:	Muhayat, M.IT
Praktikum	:	P1 – **Setup Environment & Git Workflow**
Nama Mahasiswa	:	Husna Norgina
NIM	:	230104040056
Kelas	:	TI23B
Link Repo Github	:	https://github.com/husna-norgina/bigdata-technology-2026
Tanggal Praktikum	:	19-02-2026

A. Tujuan Praktikum

1. Menggunakan VS Code sebagai environment kerja
2. Menggunakan PowerShell untuk menjalankan perintah
3. Menginstal dan menjalankan PySpark
4. Membuat cluster gratis MongoDB Atlas
5. Membuat struktur project professional
6. Menggunakan Git dan GitHub
7. Menjalankan Spark job sederhana

B. Gambaran Besar Workflow

Alur yang akan disiapkan:
VS Code
→ PowerShell
→ PySpark
→ MongoDB Atlas (Cloud)
→ Struktur Project
→ Git Versioning
→ Spark Job

C. Technology Stack

1. VS Code – Development Environment
2. PowerShell – Command Line Interface
3. Python 3.10 – Bahasa Pemrograman Utama
4. PySpark – Distributed Data Processing Engine
5. MongoDB Atlas – Cloud Database
6. Git & GitHub – Version Control & Collaboration

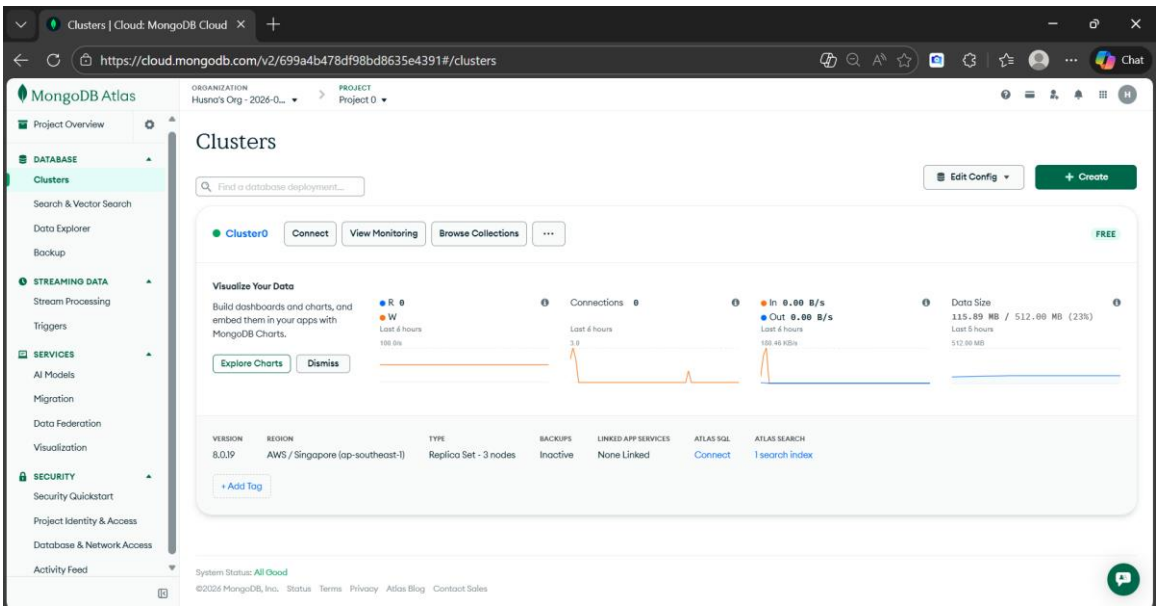
D. Output

1. Screenshot Spark Berjalan

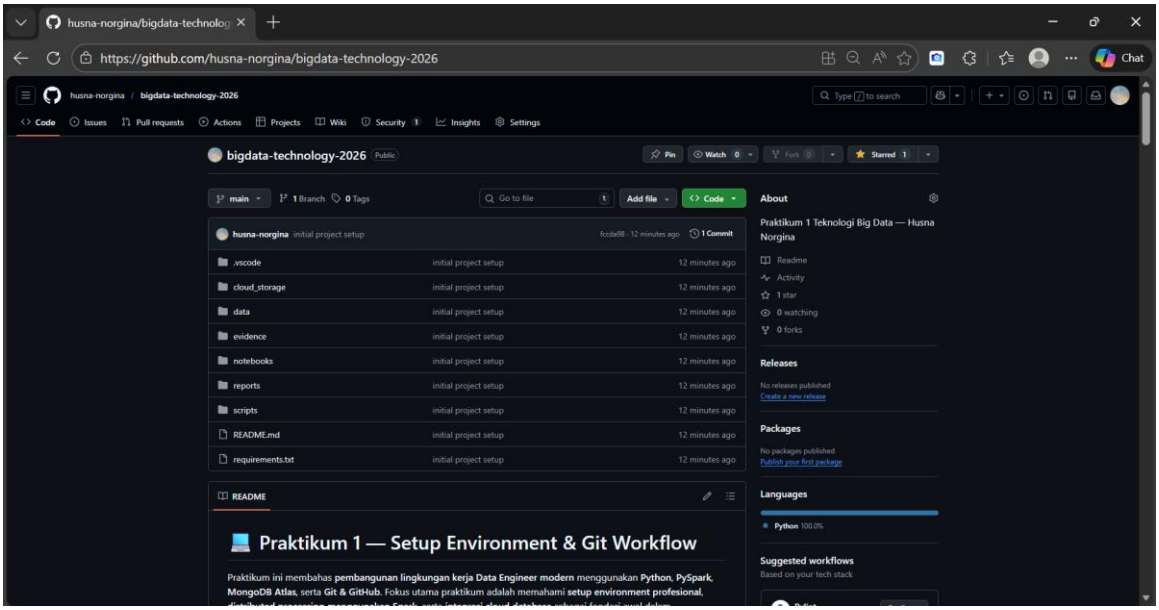
```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   powershell

PS D:\HUSNA NORGINA_KULIAH\IT_Husna Norgina_Semester 6\Teknologi Big Data\bigdata-project> python
Python 3.10.0 (tags/v3.10.0:b494f59, Oct  4 2021, 19:00:18) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> from pyspark.sql import SparkSession
>>> spark = SparkSession.builder \
... .appName("TestSpark") \
... .getOrCreate()
WARNING: Using incubator modules: jdk.incubator.vector
26/02/22 13:14:56 WARN Shell: Did not find winutils.exe: java.io.FileNotFoundException: java.io.FileNotFoundException: HAD
OOP_HOME and hadoop.home.dir are unset. -see https://cwiki.apache.org/confluence/display/HADOOP2/WindowsProblems
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/02/22 13:14:56 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java clas
ses where applicable
>>> print("Spark berhasil dijalankan!")
Spark berhasil dijalankan!
>>> exit()
PS D:\HUSNA NORGINA_KULIAH\IT_Husna Norgina_Semester 6\Teknologi Big Data\bigdata-project> SUCCESS: The process with PID 9
92 (child process of PID 24956) has been terminated.
SUCCESS: The process with PID 24956 (child process of PID 21816) has been terminated.
SUCCESS: The process with PID 21816 (child process of PID 9800) has been terminated.
                                                                          Finish Setup   Go Live
```

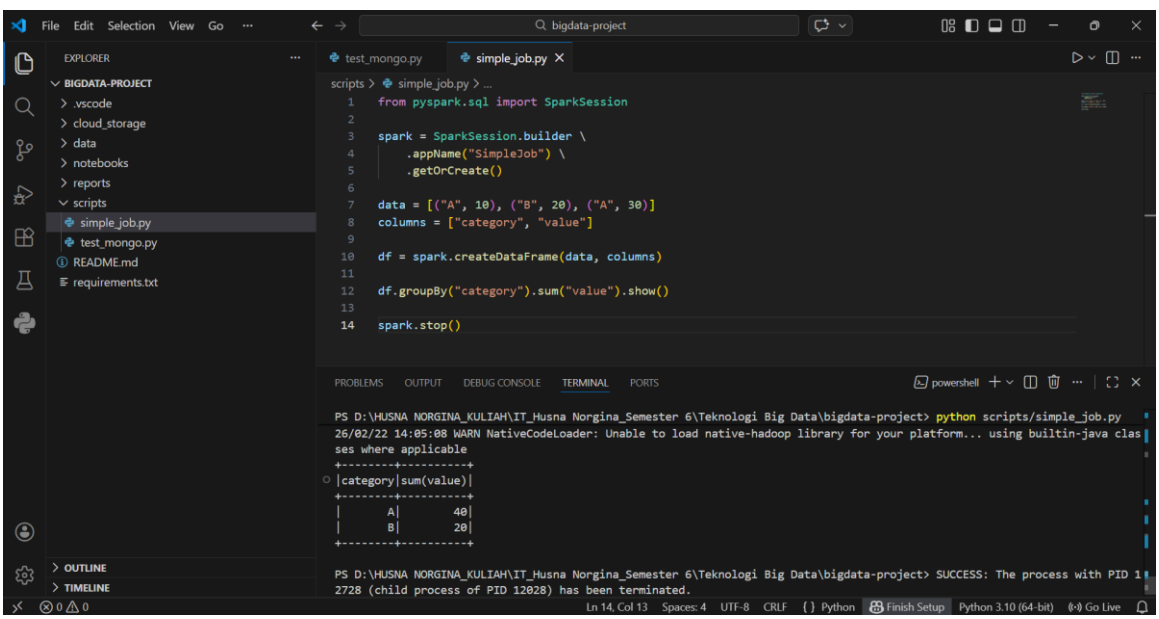
2. Screenshot MongoDB Atlas Cluster Active



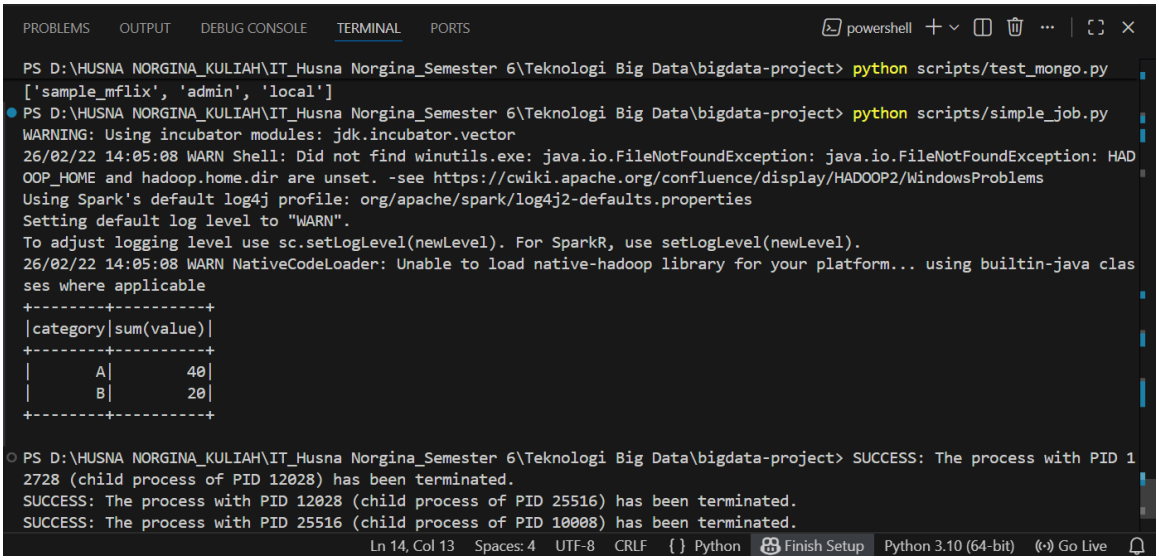
3. Screenshot Link GitHub Repository



4. Screenshot File simple_job.py



5. Screenshot Hasil Eksekusi Spark



```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                   powershell

PS D:\HUSNA NORGINA_KULIAH\IT_Husna Norgina_Semester 6\Teknologi Big Data\bigdata-project> python scripts/test_mongo.py
['sample_mflix', 'admin', 'local']
PS D:\HUSNA NORGINA_KULIAH\IT_Husna Norgina_Semester 6\Teknologi Big Data\bigdata-project> python scripts/simple_job.py
WARNING: Using incubator modules: jdk.incubator.vector
26/02/22 14:05:08 WARN Shell: Did not find winutils.exe: java.io.FileNotFoundException: java.io.FileNotFoundException: HAD
OOP_HOME and hadoop.home.dir are unset. -see https://cwiki.apache.org/confluence/display/HADOOP2/WindowsProblems
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/02/22 14:05:08 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java clas
ses where applicable
+--------+----------+
|category|sum(value)|
+--------+----------+
|       A|        40|
|       B|        20|
+--------+----------+

PS D:\HUSNA NORGINA_KULIAH\IT_Husna Norgina_Semester 6\Teknologi Big Data\bigdata-project> SUCCESS: The process with PID 1
2728 (child process of PID 12028) has been terminated.
SUCCESS: The process with PID 12028 (child process of PID 25516) has been terminated.
SUCCESS: The process with PID 25516 (child process of PID 10008) has been terminated.
Ln 14, Col 13   Spaces: 4   UTF-8   CRLF   { } Python   Finish Setup   Python 3.10 (64-bit)   Go Live
```

E. Analisis

Praktikum 1 berhasil membangun environment Big Data yang terstruktur dengan menggunakan Python, PySpark, MongoDB Atlas, serta Git dan GitHub. PySpark berjalan dengan baik melalui pembuatan SparkSession, DataFrame, dan proses agregasi data tanpa error. MongoDB Atlas sebagai cloud database juga berhasil dibuat dan terkoneksi melalui Python. Struktur project tersusun rapi sesuai standar data engineer sehingga memudahkan pengelolaan dan pengembangan lanjutan, sementara penggunaan Git dan GitHub memastikan version control berjalan sistematis. Secara keseluruhan, seluruh komponen berjalan stabil, terintegrasi dengan baik, dan telah memenuhi tujuan praktikum dalam membangun fondasi workflow Data Engineer yang profesional dan cloud-ready.

F. Kesimpulan

Berdasarkan hasil Praktikum 1 – Setup Environment & Git Workflow, dapat disimpulkan bahwa pembangunan environment kerja merupakan langkah awal yang sangat penting dalam pengembangan sistem Big Data. Melalui penggunaan PySpark sebagai distributed processing engine, MongoDB Atlas sebagai cloud database, serta Git dan GitHub sebagai version control, mahasiswa memperoleh pemahaman mengenai workflow profesional yang digunakan dalam industri data modern. Praktikum ini tidak hanya melatih kemampuan teknis instalasi dan konfigurasi, tetapi juga membangun pola pikir distributed computing, cloud integration, dan manajemen project yang terstruktur. Secara keseluruhan, praktikum ini berhasil membangun environment yang stabil, terintegrasi, dan siap digunakan sebagai fondasi untuk tahap lanjutan seperti data ingestion, ETL pipeline, data warehouse, hingga machine learning dalam ekosistem Big Data.